

ANALYSIS OF BLU-DROP (Limitations & Future Enhancement)

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Abstract—Peer-to-peer (P2P) communication systems using Bluetooth and Wi-Fi Direct enable decentralized data exchange without infrastructure. However, flooding-based communication suffers from severe scalability limitations, including broadcast storms, excessive memory consumption, and high latency due to redundant message propagation. This paper presents a comprehensive analysis of these limitations and proposes a hierarchical clustering-based routing model combined with Bloom filter-based memory optimization. Analytical modeling and numerical evaluation demonstrate that the proposed approach reduces transmission complexity from $O(N^2)$ to near-linear behavior, hop count from linear to logarithmic growth, and memory usage by up to $60\times$. The results validate the effectiveness of structured communication and probabilistic data structures for large-scale decentralized networks.

Index Terms—P2P, Bluetooth, Wi-Fi Direct, Bloom Filter, Hierarchical Clustering, Scalability.

I. INTRODUCTION

Peer-to-peer communication plays a critical role in applications such as disaster response, proximity-based sharing, and decentralized networking. Technologies such as Bluetooth Low Energy (BLE) and Wi-Fi Direct, integrated through APIs like Google Nearby Connections, enable direct communication between devices without centralized infrastructure.

Despite their advantages, these systems are primarily designed for small-scale deployments. As network size increases, flooding-based communication results in redundant message propagation, leading to broadcast storms, increased latency, high energy consumption, and excessive memory usage.

This paper addresses these challenges by:

- Analyzing limitations of flooding-based communication
- Proposing hierarchical clustering with routing optimization
- Introducing Bloom filter-based memory optimization
- Providing numerical and analytical validation of scalability improvements

II. SYSTEM MODEL & METHODOLOGY

A. Network Assumptions

- Total number of devices in the network = N
- Average number of neighbouring nodes per node = $k \approx 4$ (based on P2P_CLUSTER constraints)
- Cluster size = $c \approx 10$ (design parameter)
- Number of clusters = N/c

B. Performance Metrics

- Total Transmissions (T)
- Hop Count (H)
- Memory Consumption (M)

C. Evaluation Methodology

- Analytical modeling of communication and memory complexity
- Numerical evaluation using scalable dataset
- Comparative analysis between flooding and proposed models

III. EXISTING MODEL: FLOODING ALGORITHM

In the flooding-based model, each node rebroadcasts received messages to all its neighbors. While duplicate suppression prevents reprocessing, it does not eliminate redundant transmissions, resulting in excessive network overhead.

A. Communication Complexity Approximation

- 1) Every device broadcast message to every neighbour device.
- 2) N (Devices) $\times N$ (Transmission).
- 3) P2P Cluster strategy provide 3–5 stable connections.
- 4) Due to redundancy and overlapping of path the complexity remains $O(N^2)$.

$$T_{flood} = O(N^2) \quad (1)$$

B. Hop Count Complexity Approximation

Three possible patterns of connections can be formed:

- 1) **Tree Like Structure:** Connection of devices can be formed in a tree like structure which is ideal but very less probable to happen.
 - One root node/device.
 - k -balanced tree (k – Number of child node/connected devices).

$$H_{tree} = O(\log_k N) \quad (2)$$

- 2) **Graph like Structure:** Connection of devices form a graph like structure, there can be two possible formations.

- **Linear Chain:** When devices form connection in a chain like formation, therefore hop count becomes linear.

$$H_{chain} = O(N) \quad (3)$$

- **Random Graph:** When devices randomly connects to neighbour devices (3–5). This is the most probable case of connection in Blu-Drop. Complexity cannot be correctly evaluated due to randomness.

Approximation of complexity for Random Graph:

- Each hop adds at least one new device from the neighbours.
- Not all k neighbours are new devices added to network due to redundant path.
- New devices added per hop is $x \cdot k$ (x is connectivity factor, $x \leq 1$).
- Let H be number of hops required to deliver message to all the N devices.

So after H hops:

$$N \approx x \cdot k \cdot H \implies H \approx N/(x \cdot k) \implies H = O(N/k) \quad (4)$$

C. Memory Complexity

- 1) Size of uuid is 128 bits = 16 bytes. Converted to string due to:
 - Persistent storage prefers string based identifiers.
 - No need for serialization logic.
 - No byte conversion.
 - Direct storage and retrieval.
- 2) String consists of 36 characters:
 - 32 hex characters.
 - 4 hyphens.
- 3) Size of string = 36 bytes.
- 4) In Java/Kotlin character size is of 2 bytes.
- 5) Size of string = $36 \times 2 = 72$ bytes.

$$M_{uuid} = 72 \times N \text{ bytes} \implies M_{uuid} = O(N) \quad (5)$$

D. Limitations

- **Broadcast Storm:** Uncontrolled rebroadcasting leads to redundant transmissions and collisions.
- **Network Congestion:** Increased traffic causes packet loss and delays.
- **High Latency:** Unstructured propagation leads to large hop counts.
- **Memory Overhead:** Linear growth in stored message identifiers.
- **Energy Consumption:** Continuous forwarding drains battery resources.

IV. PROPOSED MODEL: HIERARCHICAL CLUSTERING + ROUTING

The proposed model introduces hierarchical clustering combined with structured routing and memory optimization.

A. Architecture

- Devices are grouped into clusters of size c
- Each cluster has a Cluster Head (CH)
- CHs form a higher-level communication network

B. Message Flow

Device $\rightarrow$ Cluster Head $\rightarrow$ Cluster Network $\rightarrow$ Target
Cluster $\rightarrow$ Device

C. Communication Complexity

The proposed hierarchical clustering model reduces communication complexity by dividing the network into smaller clusters. Limiting global communication to cluster heads, the proposed model significantly reduces redundant transmissions compared to flooding-based approaches, enabling efficient scalability.

- Intra-cluster communication contributes $O(N \cdot c)$.
- Inter-cluster communication among cluster heads contributes $O((N/c)^2)$.

$$T_{total} = N \cdot c + (N/c)^2 \quad (6)$$

D. Hop Count Complexity

In the proposed hierarchical model, hop count is significantly reduced due to structured communication through cluster heads. The message first reaches the cluster head, is routed across cluster heads, and finally delivered to the destination node. Since the number of clusters is N/c , the inter-cluster hop count grows logarithmically.

- Sender_device to Sender's Cluster Head (Hop = $\log_k c$).
- Number of clusters in Cluster Head Network = N/c .
- Routing avoids redundant paths so complexity will be logarithmic:
 - Number of CH neighbour, P2P Cluster supports 3–5 neighbours (k).
 - Complexity = $\log_k(N/c)$.
- Destination's Cluster Head to Destination_device (Hop = $\log_k c$).

Total hop count derivation:

$$H_{total} = 2 \cdot \log_k c + \log_k(N/c) \quad (7)$$

$$H_{total} = \log_k c^2 + \log_k(N/c) \quad (8)$$

$$H_{total} = \log_k(N \cdot c) \quad (9)$$

E. Memory Optimization

A Bloom filter stores message identifiers using a compact bit array rather than explicitly storing each identifier. When a message is received, multiple hash functions are applied to its identifier, and the corresponding bit positions are set in the array. To check for duplicates, the same hash functions are applied, and the corresponding bits are examined. If all bits are set, the message is considered a probable duplicate; otherwise, it is treated as new.

The memory requirement per element in a Bloom filter is given by:

$$c = -\frac{\ln p}{(\ln 2)^2} \quad (10)$$

where p represents the false positive probability. 72 byte per message UUID is reduced to approximately 1.2 bytes per element in bit array.

$$c = -\frac{\ln p}{(\ln 2)^2} = 9.6 \text{ bits or } 1.2 \text{ byte} \quad (11)$$

$$M = N \times c \implies M = O(N) \quad (12)$$

F. Advantages

- Reduced global communication overhead
- Logarithmic hop count growth
- Significant memory reduction
- Improved scalability and efficiency

V. RESULTS & COMPARATIVE ANALYSIS

The comparative analysis highlights significant improvements across all performance metrics.

TABLE I
PERFORMANCE METRICS COMPARISON

N	T_{Flood}	T_{Prop}	H_{Flood}	H_{Prop}	UUID (KB)	Bloom (KB)
100	10000	1100	25	5	7.03	0.12
200	40000	2400	50	6	14.06	0.23
300	90000	3900	75	6	21.09	0.35
400	160000	5600	100	6	28.12	0.47
500	250000	7500	125	7	35.16	0.59

A. Communication Improvements

- Flooding: 1,000,000 transmissions ($N = 1000$)
- Proposed: 20,000 transmissions
- Improvement: $\approx 50\times$

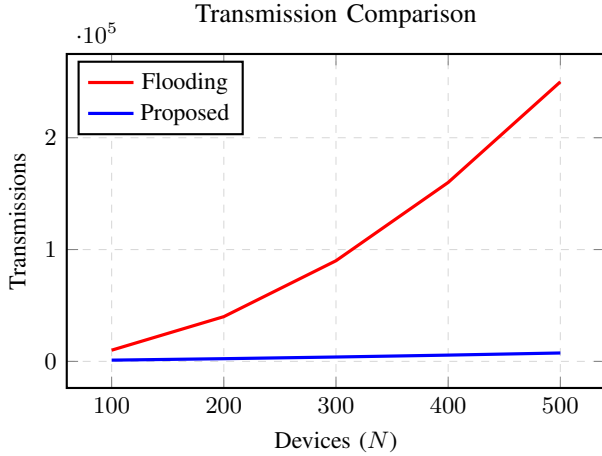


Fig. 1. Transmission Comparison: Quadratic vs near-linear complexity.

B. Latency (Hop Count) Improvements

- Flooding: ≈ 250 hops
- Proposed: ≈ 5 hops
- Improvement: $\approx 50\times$

C. Memory Usage Improvements

- UUID Storage: 7.2 MB (100,000 messages)
- Bloom Filter: 120 KB
- Improvement: $\approx 60\times$

Hop Count Comparison

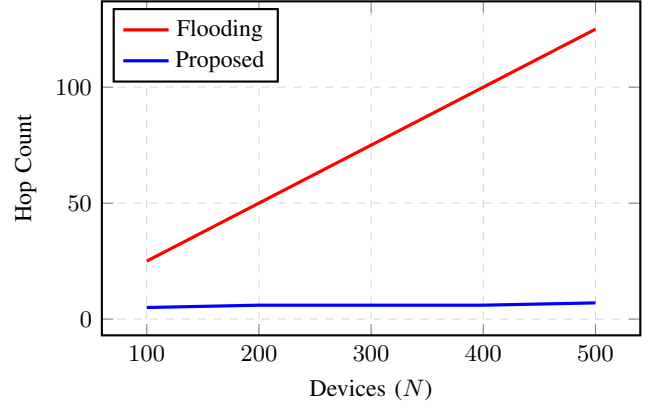


Fig. 2. Hop Count Comparison: Linear vs logarithmic reduction.

Memory Usage Comparison

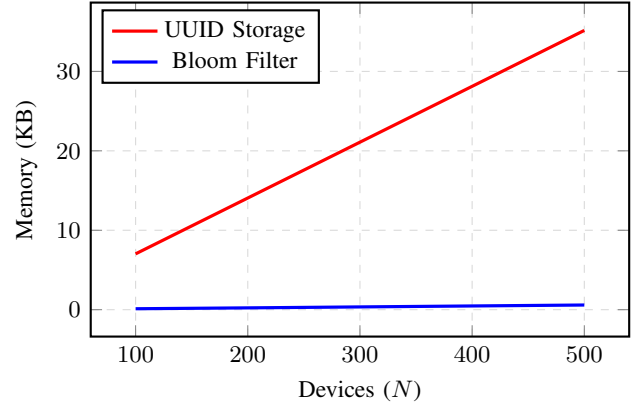


Fig. 3. Memory Usage Comparison: UUID vs Bloom Filter.

D. Observations

- Flooding exhibits quadratic growth and poor scalability
- Proposed model maintains near-linear communication and logarithmic latency
- Memory optimization significantly reduces resource consumption

VI. CONCLUSION

This paper analyzed the limitations of flooding-based peer-to-peer communication systems and proposed a hierarchical clustering-based routing model combined with Bloom filter-based memory optimization. The results demonstrate significant improvements in communication efficiency, latency reduction, and memory utilization.

By transforming unstructured communication into a structured and scalable architecture, the proposed model enables efficient operation in large-scale networks. Future work may include dynamic cluster formation, adaptive routing strategies, and real-world deployment for performance validation.

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